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Hamed Mahdavi

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Education

Ph.D. in Computer Science and Engineering **Jan 2022–Nov 2026 (expected)**
Pennsylvania State University *State College, PA*
Advisor: Prof. Vasant Honavar · Thesis: Solving Contest-Level Math Problems Using LLMs

M.Sc. in Artificial Intelligence **Sep 2017–Dec 2020**
Sharif University of Technology *Tehran*
Advisor: Prof. Hamid Beygi · Thesis: Approximate Inference in Probabilistic Network Models

B.Sc. in Computer Engineering **Sep 2012–Sep 2017**
Sharif University of Technology *Tehran*
Advisor: Prof. Hossein Sameti · Thesis: A Survey on Speech Enhancement Algorithms

Professional and Academic Experience

Research Scientist Intern **Mar 2026–Aug 2026**
Prelude *New York, NY*
- Developed agentic workflows and evaluation harnesses for open-ended creative content generation, with a focus on generating synthetic reasoning data for creativity.
- Built pipelines that transform agentic workflows into training data, studying the interaction between mid-training and post-training for creative reasoning capabilities.
- Trained reasoning and non-reasoning reward models for creative judgment and led data curation for reward modeling in the creative judgment domain.

Collaborator, Gemini IMO 2025 Project **Summer 2025**
Google *New York, NY*
- Provided high-quality verifiable data and evaluations to the Gemini Data team under the supervision of Dr. Vahab Mirrokni, resulting in a gold medal at the International Mathematical Olympiad (IMO).

Research Assistant **Jan 2022–Present**
Pennsylvania State University *State College, PA*
- Design agentic workflows for math problem solving and high quality mathematical data generation; build and run evaluation harnesses for contest-level math (benchmarks, metrics, ablations, failure analysis); develop GNN models for interatomic potentials in materials science.

Data Scientist Intern **Jun 2023–Aug 2023**
Arconic Technology Center *New Kensington, PA*
- Worked in the Fabrication Technology group under the supervision of Martin Marinack; tackled several data-driven optimization problems to improve Arconic's fabrication processes (welding, rolling, and casting).

Research Intern **Aug 2018–Jan 2019**
Max Planck Institute for Software Systems *Saarbrücken, Germany*
- Worked on fundamental aspects of reinforcement learning, focusing on multi-agent, adversarial, and human-centered reinforcement learning.
- Studied how to design an AI agent that can robustly cooperate with agents of unknown types. The project was inspired by real-world applications in which an AI agent must cooperate with new users or agent types encountered only after deployment.

Software Engineering Intern **Jul 2017–Sep 2017**
Faragostar *Tehran*
- Developed a compression algorithm that allowed Faragostar to retain its official documents while minimizing storage requirements.
- Configured infrastructure for Elasticsearch, enabling more efficient and complex searches in the company's app compared to the previous SQL-based search tool.

Mathematical Olympiad Teacher **Oct 2012–Apr 2013**
Allameh Helli High School *Tehran*
• Taught number theory to academically gifted students at Allameh Helli High School.

Honors and Awards

Received $100k USD Unrestricted Research Gift from Google **May 2025**
Research Funding
• Awarded to support research on enhancing large language models for mathematics through synthetic data generation and agentic workflows for math problem solving.

Top National Rankings in M.Sc. Entrance Exam **Jun 2017**
Nationwide Ranking
• Placed 5th (AI), 2nd (Networks and Security), and 4th (IT) nationwide in a highly competitive M.Sc. entrance exam.

Ranked 217th in National University Entrance Exam **Jun 2012**
Nationwide Ranking
• Ranked in the top 0.1% among 260,000+ examinees nationwide.

Silver Medal in the 29th National Mathematical Olympiad **Sep 2011**
National Competition
• Selected as one of 40 participants from 100,000+ candidates; awarded Silver Medal for outstanding performance in the National Mathematical Olympiad.

Selected Publications

• **Splicing Hints into Reasoning Traces: Distillation Beyond What the Demonstrator Can Solve** — *arXiv:2610.13996, Submitted to ICLR* **2027**
H. Mahdavi, P. Mahdavinia, M. Farhadi, V. Honavar

• **RefGrader: Automated Grading of Mathematical Competition Proofs using Agentic Workflows** — *Math-AI Workshop at NeurIPS* **2025**
H. Mahdavi, A. Hashemi, M. Daliri, P. Mohammadipour, A. Farhadi, S. Malek, Y. Yazdanifard, N. Mireshghallah, A. Khasahmadi, V. Honavar

• **CombiGraph-Vis: A Multimodal Olympiad Benchmark for Discrete Mathematical Reasoning** — *Math-AI Workshop at NeurIPS* **2025**
H. Mahdavi, A. Hashemi, M. Daliri, P. Mohammadipour, A. Farhadi, S. Malek, Y. Yazdanifard, A. Khasahmadi, V. Honavar

• **Brains vs. bytes: Evaluating LLM Proficiency in Olympiad Mathematics** — *COLM Conference* **2025**
H. Mahdavi, A. Hashemi, M. Daliri, P. Mohammadipour, A. Farhadi, S. Malek, Y. Yazdanifard, A. Khasahmadi, V. Honavar

• **Beyond Training Data: How Elemental Features Enhance ML-Based Formation Energy Predictions** — *Digital Discovery* **2025**
H. Mahdavi, V. Honavar, D. Morgan

• **Towards deployment of robust cooperative AI agents: An algorithmic framework for learning adaptive policies** — *AAMAS Conference* **2020**
A. Ghosh, S. Tschiatschek, H. Mahdavi, A. Singla

Skills

Programming	Python, Java, C#, Scala, LaTeX, C
ML Tools	PyTorch, HuggingFace, vLLM, Verl, Verifiers, NeMo Framework, NeMo-RL, NeMo Auto-Model, scikit-learn, NumPy, SciPy, Pandas
LLM Skills	Evaluation, Agentic Workflows, Agentic Loop Engineering, Synthetic Data Generation, Data Curation
Development	Cursor, Git, Python Django, Play Framework, SQL, Bootstrap
Communication	English (fluent)